

Root Cause Analysis — RCA-2025-003

WIN-10130 Chronic Failure Pattern (CAISO-WIND-026)

Field	Value
Document ID	RCA-2025-003
Date Issued	2026-02-04
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Region	CAISO (with NW-PowerPool sister-fleet implications)
Site	CAISO-WIND-026
Asset	WIN-10130 (Siemens Gamesa, wind_turbine)
Failure Mode Classification	M-3 (Mechanical Cumulative) + V-1 (Vendor Governance) + P-2 (Policy Conflict)
Distribution	CAISO Operations, NW-PowerPool Reliability Engineering, OEM Liaison, Vendor Management, CFO Office, Executive Operations Committee

Executive Summary

WIN-10130 at CAISO-WIND-026 logged **14 forced outages** and **20,494 MWh** of lost generation in FY2025 — the worst single-asset outcome in the Cascadia fleet. The event detail is captured in OIR-2025-010.

The root cause is **a policy-conflict deadlock between Siemens Gamesa OEM Manual Rev. 4.2 and Cascadia internal Standard MS-2025-03**, with **MOC-2025-022 the unresolved instrument intended to break the deadlock**. The deadlock has now persisted for over 10 months. During that time WIN-10130 has been repair-cycled 14 times, costing Cascadia approximately \$2.6M in vendor invoices and producing the largest revenue loss on any single asset in FY2025.

The WIN-10130 pattern is not isolated. Three NW-PowerPool sister-units (WIN-10131, WIN-10011, WIN-10054) show similar trend slopes and are governed by the same policy conflict. A coordinated FY26 capital plan is the appropriate response.

1. Failure Mode Classification

M-3 (Mechanical Cumulative): Across FY2025, WIN-10130 displayed continuous degradation rather than discrete failures. Iron particulate in gearbox oil rose from 6 ppm (Jan) to 14 ppm (Oct). Vibration RMS rose from 5.2 mm/s (Jan baseline) to 7.4 mm/s (Oct peak). Heat-exchanger fouling increased over the year. Each repair cycle has restored the unit to within a narrowing window of pre-failure state.

V-1 (Vendor Governance): Three different field service contractors handled the 14 events (CPS, Apex, Heartland). Service history is fragmented; no single vendor has a coherent view of the asset's condition. Repair recommendations have been inconsistent. Re-baselining occurs at each vendor handoff.

P-2 (Policy Conflict): Siemens Gamesa OEM Manual Rev. 4.2 specifies replacement at vibration RMS 6.0 mm/s on the high-speed shaft. Cascadia Standard MS-2025-03 sets the threshold at 8.0 mm/s. WIN-10130 operated above the OEM threshold for **178 cumulative hours** in FY2025 without replacement. The OEM has issued a \$1.2M gearbox replacement quote (Q-SG-25-0211, issued 2025-04-30). The quote has been rejected pending MOC-2025-022 resolution; MOC-2025-022 has been pending engineering review since 2025-03-04.

2. Fishbone / Contributing Factors

Method:

- MOC-2025-022 is the unresolved instrument. It was opened 2025-03-04. As of report date (2026-02-04), 11 months elapsed. The MOC has been routed three times for engineering review; each routing has stalled. The proximate cause of each stall is unclear and is a governance concern.
- Standard MS-2025-03 was issued February 2025 with thresholds 30% above the prior internal standard (which matched the OEM). The decision to raise the thresholds was supported by an internal study referenced in MS-2025-03 §2 but the study is not attached to the MOC.

Machine:

- WIN-10130 has been in service 4.3 years (commission date 2021-09). 12 sister-units in the same product line are deployed across CAISO-WIND-026 (4 units) and NW-PowerPool-WIND-024 (8 units). Mean service age across the population: 3.8 years.
- The product line was the first Siemens Gamesa generation to ship with the upgraded high-speed-shaft bearing assembly. OEM monitoring data (shared under

VRA-2025-001) indicates that the upgraded bearing requires the lower vibration threshold cited in Rev. 4.2.

Money:

- Apex invoiced \$1.91M against this asset in FY2025 (6 events). Average per-event \$318K. Three Apex recommendations to replace components were rejected; the field repairs that followed were ~25% of quoted cost.
- The \$1.2M gearbox replacement quote has been outstanding since April 2025. Had it been executed in May 2025, the avoided revenue-loss in H2 FY25 would have been approximately \$740K (Aug-Dec events on this asset, 6 events × ~\$120K each).

Materials:

- The Cascadia OEM service-bulletin intake captured Siemens Gamesa bulletin SG-PB-2025-02 (April 2025) recommending pitch accumulator pre-charge verification every 6 months. The bulletin is implicated in the WIN-10011 December event (OIR-2025-009) for the sister-fleet at NW-PowerPool-WIND-024.

3. Timeline of Relevant CARs and Outages

Date	Event	Reference
2025-01-20	WIN-10130 first FY25 outage	(nw_powerpool_outages roll-up; OIR-2025-010)
2025-02	Standard MS-2025-03 issued	MS-2025-03
2025-03-04	MOC-2025-022 opened	MOC-2025-022
2025-03-09	WIN-10130 overheating event, 213 h	OIR-2025-010 / OUT-2025-0556
2025-04-30	Siemens Gamesa quote Q-SG-25-0211 issued	\$1.2M gearbox
2025-05-16	Vibration_excess event, 211 h	OUT-2025-0566
2025-06-21	Vibration_excess event, 202 h	OUT-2025-0558
2025-07-18	RCA-2025-001 issued (Spring Vibration Cascade)	RCA-2025-001
2025-10-10	Vibration_excess event, 183 h	OUT-2025-0559
2025-12-07	Final FY25 event on WIN-10130 (pitch system)	(OIR-2025-010 aggregate)
2026-01-09	OIR-2025-010 issued (multi-event roll-up)	OIR-2025-010

Date	Event	Reference
2026-02-04	This RCA issued	RCA-2025-003

4. Quantified Business Impact

Item	Value (USD)
FY25 Apex invoicing on WIN-10130	\$1,910,000
FY25 CPS invoicing on WIN-10130	\$312,000
FY25 Heartland invoicing on WIN-10130	\$284,000
FY25 Siemens Gamesa OEM warranty review	\$112,000
FY25 total vendor cost on WIN-10130	\$2,618,000
FY25 estimated revenue loss (20,494 MWh × \$60)	\$1,229,665
FY25 total impact (single asset)	~\$3.85M
Siemens Gamesa gearbox replacement quote (alternative)	\$1,200,000
Net cost differential of repair-cycle vs replacement	~\$1.42M (excluding revenue loss)
Net cost differential including revenue loss	~\$2.65M

The economic decision is unambiguous. Replacing the gearbox would have cost less than continued repair-cycling, even before counting lost generation revenue. The deadlock between OEM Manual Rev. 4.2 and Standard MS-2025-03 is materially costing Cascadia roughly \$2.65M per affected asset per year.

For the 12-unit sister-fleet, applying the same arithmetic: gross potential impact ~\$31.8M annually. Even discounting heavily for asset variability, the FY26 capital case is compelling.

5. Long-Term Corrective Recommendations

1. **Force resolution of MOC-2025-022 within 30 days.** Two outcomes are acceptable:
2. (a) Adopt OEM Manual Rev. 4.2 threshold of 6.0 mm/s. Budget \$4.8M for gearbox replacement on the 4 most-affected sister-units (WIN-10130, -10131, -10011, -10054) in FY26 Q1.

3. (b) Formally adopt MS-2025-03 threshold of 8.0 mm/s as controlling, void the Siemens Gamesa replacement quote, and update the MSA to disable OEM replacement-quote workflow.
The current limbo is the worst outcome.
4. **Single-prime vendor at CAISO-WIND-026** for the next 12 months. Recommend either CPS or Heartland (not Apex). Re-establish coherent service history. Refer to VRA-2025-006.
5. **FY26 capital planning workstream** for the 12-unit sister-fleet across CAISO-WIND-026 and NW-PowerPool-WIND-024. Cross-reference RAM-2025-007.
6. **OEM engagement formalization.** Open a quarterly OEM technical review with Siemens Gamesa to surface their monitoring data and reconcile with internal vibration baselines. Refer to VRA-2025-001.
7. Add WIN-10130, -10131, -10011, -10054 to a single asset-watch list governed by this RCA's corrective plan.

6. Vendor Performance Commentary

Siemens Gamesa OEM has applied Manual Rev. 4.2 consistently. The OEM is not behaving unreasonably; they are applying their published manual. The conflict is internal to Cascadia.

Apex Emergency Grid Solutions has invoiced \$1.91M against this asset in FY2025 (~73% of total vendor cost on WIN-10130). Apex's repeated replacement recommendations on this asset's components have been the primary driver of internal engineering counter-proposals. The pattern matches the Period A and Period B observations in RCA-2025-001 and RCA-2025-002.

Heartland Reliability Partners has handled three events on this asset at lower cost than Apex with comparable outcome quality. Heartland's performance-based contract proposal (referenced in narrative bible §2.7) should be re-examined for CAISO operations.

Cascadia Power Services Inc. has handled four events on this asset at lowest unit cost. CPS lacks the depot footprint to be a single-prime at CAISO; investment in a Bakersfield expansion would be required if single-prime CPS is selected.

Signatures

Role	Name	Date
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Role	Name	Date
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Distribution: CAISO Operations · NW-PowerPool Reliability Engineering · OEM Liaison · Vendor Management · CFO Office · Executive Operations Committee

References: OIR-2025-002 · OIR-2025-009 · OIR-2025-010 · MOC-2025-022 · RAM-2025-007 (draft) · VRA-2025-001 · VRA-2025-006 · EMR-2025-002 · RCA-2025-001 · RCA-2025-002 · Siemens Gamesa OEM Manual Rev. 4.2 · Quote Q-SG-25-0211 · Standard MS-2025-03 · Bulletin SG-PB-2025-02